

Coffee Shop Website Information System

System Architecture and System Design

1. An architecture style that I use to design my coffee shop website is the component system. My thought process is breaking these big tasks into smaller components that I can reusable, I can easily replace what I've been implementing previously, for example, instead of having to create extra button what I will do is bring back the button feature I have already code and then update its logic. Another thing I have observed is I specifically customize the website in this way, rather than letting the user have to manually switch in between different .html file like manager dashboard and employee and then the customer view, I decide to wrap that component to those specific kinds of user

Manager -- has manager role

Employee -- has employee role

Customer -- has customer role

Role assign makes things clear for those components that are design to shows and specific commands this user can access to.

2. The server connects and communicates with the client through HTTP (hyper text transfer protocol locally) what it does is the front end sends to backend port and receives the information they need to send back from the back end

3. My server is host using python programming specifically I use fastAPI, build using uvicorn , and then in front end I use axios to bridge the connection. While design the system, one thing I love about using fastAPI is it very fast and the cool part of using uvicorn is whenever I make changes it reloads the server in real time (run via commandline)

4. I use MariaDb to store important user information

Network type:	MariaDB or MySQL (TCP/IP) ▼
Library:	libmariadb.dll ▼
Hostname / IP:	127.0.0.1
	☐ Prompt for credentials
	☐ Use Windows authentication
User:	root
Password:	
Port:	3306 ▲▼
	☐ Compressed client/server protocol
Databases:	Separated by semicolon ▼
Comment:	

Below are the tables I have created: KiB denotes Kilobytes

item table is used for storing coffee items

notification table is used for notification

orderitem is used for tracking order id, item id and the how much item selected

orders keep track of each transactions

user is used for tracking members

▼ coffee_shop_db 176.0 KiB	
item	16.0 KiB
notification	32.0 KiB
orderitem	48.0 KiB
orders	48.0 KiB
user	32.0 KiB

Three essential tables

notification table

Whos the sender ?

What message are they sending ?

Are they read the notification or not ? (to track they read or not tracks two things : are they online or not and two the time when the notification appears and when it close by user)
special detail : if user have not read it , shows a badge (font white , red background with number shows how many notifications are there, they can tap to view the message)

Yellow key: Primary key – unique

Green key: Foreign key userID is reference from the user table (below)

	#	Name	Datatype	Length/Set	Unsigned	Allow N...	Zerofill	Default
	1	id	INT	11	☐	☐	☐	No default
	2	userID	INT	11	☐	☐	☐	No default
	3	message	VARCHAR	255	☐	☐	☐	No default
	4	read	CHAR	50	☐	☐	☐	No default
	5	created_at	DATETIME		☐	☐	☐	current_timestam...

User table

Ah we got a red key, this is unique, this means each person email should be different

I use created_at with date time to time stamp (to debug and log in system)

	#	Name	Datatype	Length/Set	Unsigned	Allow N...	Zerofill	Default	Comment
	1	id	INT	11	☐	☐	☐	AUTO_INCREME...	
	2	name	VARCHAR	255	☐	☐	☐	No default	
	3	email	VARCHAR	255	☐	☐	☐	No default	
	4	password	VARCHAR	255	☐	☐	☐	No default	
	5	role	VARCHAR	50	☐	☐	☐	'customer'	
	6	created_at	DATETIME		☐	☐	☐	No default	

Here is what the user table store (database)

#	1 id 🔑	2 name	3 email 🔑	4 password	5 role ▲	6 created_at ▲
1	19	Olzhas Alton	olzhas@gmail.com	12345	customer	2025-10-24 16:29:37
2	20	Erik Alvin	erik@gmail.com	112233	customer	2025-10-24 16:29:37
3	21	Rahmesha Wilson	rahmesha@gmail.com	12345678	customer	2025-10-24 16:29:37
4	31	AdrielHandsome	adrielnguyenminh@gmail.com	12345	customer	2025-10-28 03:43:52
5	32	test	test@gmail.com	123	customer	2025-11-02 11:51:40
6	33	scoobydoo	scoobydoo@gmail.com	123	customer	2025-11-02 12:03:02

Here you can see all id are unique

Name from 1 – 3 are seeded

4 to 6 use for test case

So what does the database responsible for? It is a back bone of this project, where it store

1. User membership points
2. User personal information (real life it will be privacy information)
3. Bank details – cvv, 16 numbers on the card, card holder name

This application runs locally on one machine since it does not host using cloud service.

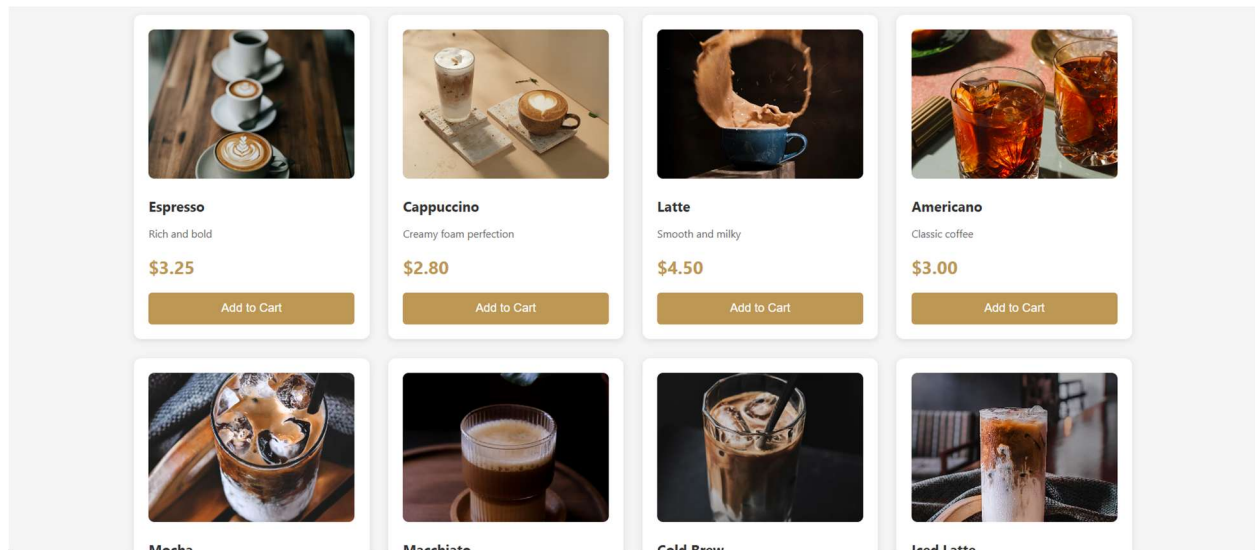
7. What does python play a role as a backend service in between database and the frontend? Python acts as an endpoint of the API get ways that use fastAPI with CRUD operations to transfer client front end interaction to process information from the database.

8. Where is the UI? At first, halfway through the project, I suddenly noticed from the mcd I couldn't install React from vite because I'm trying to use shadcnUI components, which is very nice. So, I decided to use html, css and java script. Main features are render through javascript, css , python, fastAPI and database. Html keeps minimal

Identify Subsystems

Discussion: At first glance, I thought what if I store user information on front end local storage, this in fact is a cool idea, but turns out to be very dangerous and vulnerable.

An illustration from my front-end UI menu (user, employee, and manager can see)
(note: image are references and attributes in a reference note, in the project main
readme.md images are copyright free)

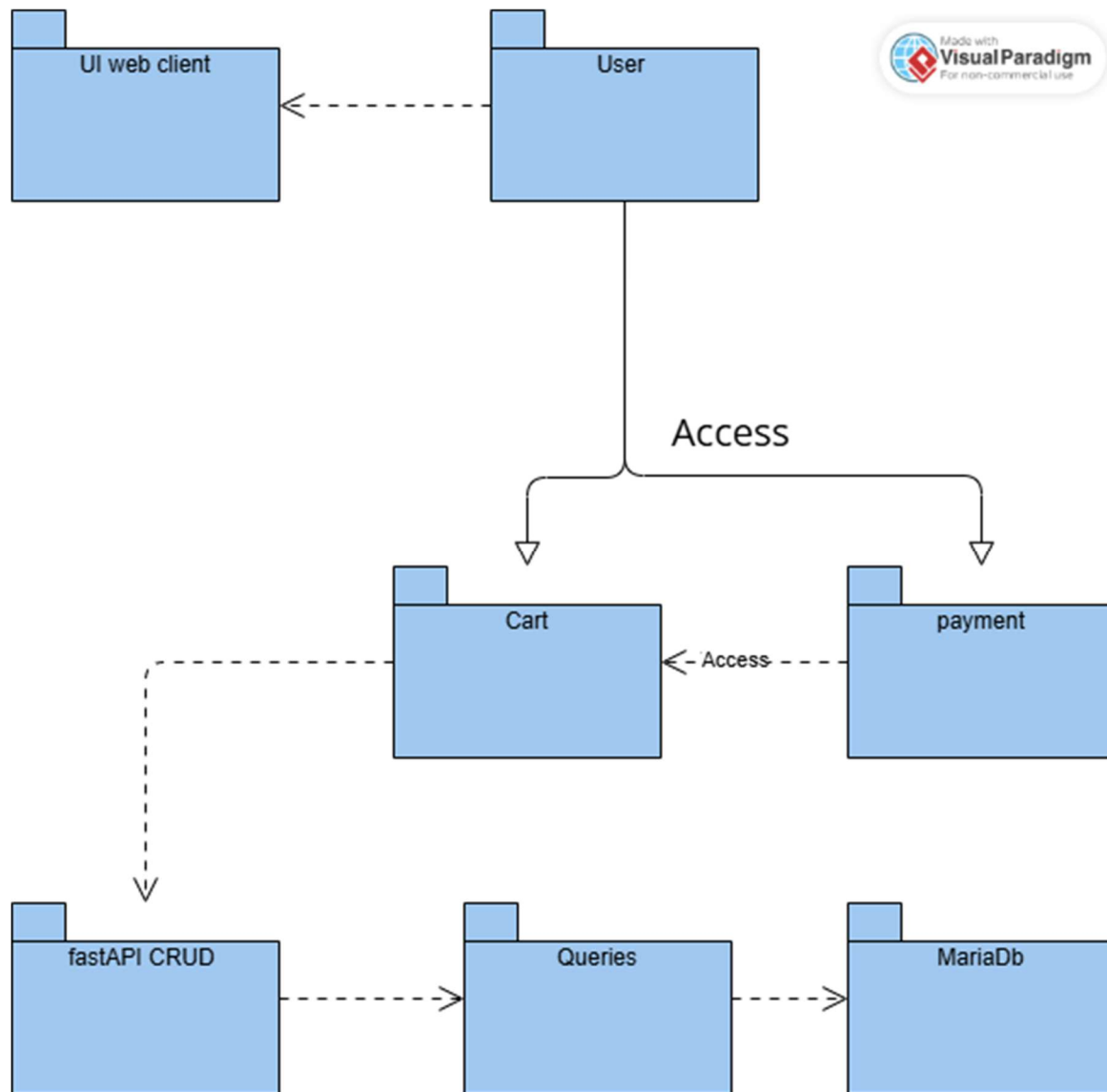


Html is use to gather user information, user type into the form label, it then validates first part by the javascript to ensure that everything is pass through first layer (8 characters, has @ in email things like that, then the information transfer by the system to the second layer which is python, it then use a class of model data, these are the user class which acts as object relation model that convert the user information into something the database can store, update , retrieves and delete through CRUD operations. The purpose is it makes life easier and efficient, how? instead of manually writing on a paper, now we use queries to makes task easier and store things digitally. A simple query can get the right information quickly within 0.12s and it is very detailed.

Add items to menu, update item from client front end to back end (real time)

SQL query is using a parameterized, which in mariadb this is very useful and a strong resistance against cross site scripting.

UML



Database Storage Layer

Data storage

What to store specifically,

1. Store dynamic information, for example how much does the user pay for an item (a drink or cake)
May be today drink cost: \$1.25 but tomorrow its \$1.5. Database management system can do this efficiently
2. Credit card: information can be store for returned purchases, what if the user needs to buy the item again, if without MariaDB, it will have to store somewhere in the back end, not secure and fast queries
3. Memberships and barcode – how much points do user have now and what's the barcode that represents the transaction that the user buy

It needs to be in a relational database with integrity check, in one of the databases, I must set the validation of quantity cannot be less than 0 because how can we order -1 items? so this is an example of one of the checks (others are referential, null check, integrity check)

Execution Orders

User interacts with the UI components at the front end, the system reacts by showing notifications, pop up dialog to ask for submit information, it then send to backend. How it glues together is when for example, if a user has tried to access let say gain access into the membership and user account without login, the fastAPI gateway and the server acts as not letting unauthorized individuals enter the frontend unless they register and login. This kind of action gives the user the freedom and ability to interact with the data visually, like primary and secondary color and gain access to role base thing they can do, like customer can only permit to view the menu, they are not allowed to have modifications on it.

What kind of system?

Time Dependency

Has time – but it displays from back end, the front end fetches as json and then displays the information (display local time, all another time stamps are track separately in database)

Major things to work on are fixing bugs and simplifying architecture (I'm quite over engineer it – it does work by connecting front end to backend, can send notification, can have role base login, can create , update, delete and get information from back end send and use by front end or backend – python)

Ah one more thing the front end have to access CORS to be able to connect to the backend, other wise it will not listen to the front end at different port.

Hardware requirements

Operating system: Window / Linux / MacOS (as long as can install dependencies to local computer)

Color: minimum 1366 x 768 or full HD 1080pixel

Computer: Home laptop

Memory: 2 Gb Ram

Hard Drive needs around 1 Gb Ram